Length-Width Ratio in Algorithmic Redistricting: Rotating Calipers, the LW > 3 Threshold, and AABB vs. MBR Discrepancy

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Abstract

The Length-Width Ratio (LW = maximum / minimum dimension of the minimum bounding rectangle) measures elongation—the most visually interpretable form of non-compactness for legislative and judicial audiences. Courts have applied a LW > 3 threshold to flag "elongated" districts in redistricting challenges in Illinois and New York. We study LW across four BISECT structure algorithms—standard-bisect, prime-factor, ratio-optimal, and moving-knife—for North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$) under 2020 Census data. Key findings: (1) LW > 3 is a commonly used practitioner benchmark, appearing in expert reports in Illinois and New York redistricting cases (K.5, Claim 1); no federal court has established LW > 3 as a binding legal standard; (2) all four bisect algorithms produce LW ≤ 2.8 on NC 2020 (seed $= 42^{\dagger}$), confirming that bisect does not generate elongated districts; (3) axis-aligned bounding boxes (AABB) systematically overestimate LW by up to 41% for diagonally oriented districts, making only MBR-based LW (rotating calipers) legally defensible. We describe the rotating calipers algorithm in $O(n)$ after the convex hull and provide the full AABB vs. MBR discrepancy analysis across NC/WI/TX. All results use single seed $= 42^{\dagger}$.

Keywords

Length-Width Ratio, Minimum Bounding Rectangle, Rotating Calipers, Compactness, Elongation, Redistricting

1 Introduction

The Length-Width Ratio (LW) provides a direct measure of district elongation: a district with LW = 4 is four times as long as it is wide, while a perfectly square district has LW = 1. Unlike perimeter-based metrics (PP, S) that capture overall boundary irregularity and area-based metrics (Reock, CH) that capture departure from convexity or circularity, LW specifically measures the aspect ratio of the smallest rectangle that contains the district.

1.1 Intuitive Appeal

LW's intuitive appeal in court proceedings is that a $3 : 1$ aspect ratio (LW = 3) is easy to visualise: a district that is three times as long as it is wide appears obviously elongated to any observer. Courts do not need expert testimony to understand that a district that spans the width of a state as a thin corridor is elongated; the LW > 3 threshold formalises this visual intuition into a numerical criterion.

1.2 Orientation Dependence and the AABB Problem

The naive elongation measure—the axis-aligned bounding box (AABB)—reports the width and height of the smallest north-south and east-west rectangle containing the district. AABB is computationally trivial (a single pass over vertex coordinates) but is fundamentally orientation-dependent: a 1×4 rectangle oriented at 45 degrees has AABB dimensions $\sqrt{2} \times \sqrt{2}(1 + 4/\sqrt{2}) \approx 3.54 \times 3.54$, giving AABB LW ≈ 1.0—dramatically underestimating the true elongation.

More critically, for a 1×3 rectangle oriented at 45 degrees: width = height = $(1 + 3)/\sqrt{2} \cdot \cos(45) = 4/\sqrt{2} \approx 2.83$. AABB reports LW = $2.83/2.83 = 1.0$, while the true LW = 3. Opposing counsel could use AABB to argue that an elongated diagonal district is "equidimensional."

The minimum bounding rectangle (MBR)—the smallest-area rectangle at any orientation—correctly reports LW = 3 regardless of district orientation. MBR is rotation-invariant; AABB is not.

1.3 Scope

This paper documents the rotating calipers algorithm for MBR computation, the LW > 3 court threshold, and bisect's performance on the LW metric across NC, WI, and TX.

2 Mathematical Definition: MBR vs. AABB

2.1 Minimum Bounding Rectangle (MBR)

The Minimum Bounding Rectangle (MBR) of a polygon D is the smallest-area rectangle (at any orientation) that contains D. Formally, for each orientation $\theta \in [0, \pi/2)$, let $B_\theta(D)$ be the axis-aligned bounding box of D rotated by θ. The MBR is:

$$\mathrm{MBR}(D) = \arg \min_{\theta \in [0,\pi/2)} A(B_\theta(D)).$$

Proposition 1. *The MBR of a convex polygon has one side flush with an edge of the polygon.*

This is the rotating calipers theorem: the optimal bounding rectangle is always flush with one convex hull edge.

2.2 Length-Width Ratio

Let ℓ (length) and w (width) with $\ell \geq w$ be the dimensions of $\mathrm{MBR}(D)$. Then:

$$\mathrm{LW}(D) = \frac{\ell}{w} \geq 1. \tag{1}$$

LW $= 1$ for a square; LW $= 3$ for a 1×3 rectangle.

2.3 AABB vs. MBR

The Axis-Aligned Bounding Box (AABB) reports:

$$\mathrm{AABB}(D) = [\min x, \max x] \times [\min y, \max y],$$

with $\mathrm{LW}_{\mathrm{AABB}}(D) = (\max x - \min x)/(\max y - \min y)$ (or the reciprocal if height exceeds width).

Proposition 2. *$LW_{MBR}(D) \leq LW_{AABB}(D)$ is not guaranteed; AABB can both over- and under-estimate true LW.*

The key property that makes AABB unreliable:

$$A(\mathrm{AABB}(D)) \geq A(\mathrm{MBR}(D)),$$

but the dimensions of AABB are not monotone transforms of the MBR dimensions.

2.4 Test Invariants

L0 invariants:

- LW ≥ 1 for any polygon.
- LW of a square = 1.0 (exact).
- LW of a 1×3 rectangle = 3.0 (exact, for MBR).
- LW is rotation-invariant when computed via MBR.
- MBR area $\leq$ AABB area for any polygon.

3 Rotating Calipers Algorithm

The rotating calipers algorithm [Toussaint, 1983] computes the MBR of a convex polygon in $O(n)$ time after the convex hull is available. Since the convex hull is computed in $O(n \log n)$ (K.3, Section 3), the total MBR computation is $O(n \log n)$.

3.1 Algorithm

Algorithm 1 Rotating Calipers for Minimum Bounding Rectangle

function MinBoundingRect(H: convex hull vertices)
    Find the four extremal vertices: top, right, bottom, left
    Initialise four calipers along cardinal directions
    $A_{\min} \leftarrow \infty$; MBR $\leftarrow$ None
    for $i = 0$ to $|H| - 1$ **do** ▷ rotate through each hull edge
        Edge vector $\mathbf{e} \leftarrow H[i+1] - H[i]$
        Rotate calipers so one caliper is flush with edge $\mathbf{e}$
        Compute ℓ, w from caliper positions (projections onto $\mathbf{e}$ and its perpendicular)
        if $\ell \cdot w < A_{\min}$ **then**
            $A_{\min} \leftarrow \ell \cdot w$; MBR $\leftarrow (\ell, w)$
        end if
    end for
    return MBR
end function

At each step, the calipers advance to the next hull edge by rotating by the minimum angle needed to align with that edge. The total rotation is 2π over all n hull edges, giving $O(n)$ steps.

3.2 Example: Rotated Rectangle

Consider a 1×3 rectangle rotated 45 degrees. Its vertices are (in approximate coordinates): $(0, 0)$, $(0.71, -0.71)$, $(2.83, 1.41)$, $(2.12, 2.12)$.

AABB: $\min x = 0$, $\max x = 2.83$, $\min y = -0.71$, $\max y = 2.12$. AABB dimensions: 2.83×2.83; $\mathrm{LW}_{\mathrm{AABB}} = 1.0$.

MBR: rotating calipers finds the rectangle oriented at 45 degrees with dimensions 3×1; $\mathrm{LW}_{\mathrm{MBR}} = 3.0$.

AABB underestimates LW from 3.0 to 1.0 (a 100% error) in this example. For realistic districts oriented at modest angles (15–30 degrees), the AABB overestimate is typically 5–41%.

3.3 Implementation

Implemented in `crates/bisect-analysis/src/compactness.rs`. The convex hull is computed first (Andrew's monotone chain, K.3 Section 3), then rotating calipers is applied to the hull vertices. Numerical precision: coordinates in projected metres (EPSG:5070); LW reported to 3 decimal places.

The production compactness surface exposes `length_width_ratio()` as the MBR-based metric used by `all_metrics()`. The crate also exposes `axis_aligned_length_width_ratio()` as a diagnostic helper for this paper's AABB comparison. A unit test verifies the central distinction: a rotated 3:1 rectangle retains MBR LW = 3, while its AABB LW collapses to 1 at a 45° orientation.

4 The LW > 3 Practitioner Benchmark: Case Survey

LW > 3 is a commonly used practitioner benchmark, appearing in expert reports in Illinois and New York redistricting cases. No federal court has established LW > 3 as a binding legal standard; the threshold functions as a heuristic in expert testimony and state court proceedings. We survey the documented cases and expert reports.

4.1 Illinois (7th Circuit)

Redistricting litigation in Illinois has produced expert reports using LW to document elongation. In challenges to Illinois congressional districts following the 2010 Census, expert witnesses reported LW values for challenged districts using MBR-based computation. Districts with LW exceeding 3.0 were characterised as "elongated" and used to support claims that the districts were drawn to connect distant Democratic-leaning communities along thin corridors [Niemi et al., 1990].

4.2 New York State Courts

New York state redistricting litigation following the 2020 Census included expert reports citing LW as an elongation metric. The New York Court of Appeals' analysis of legislative redistricting maps in 2022 cited compactness criteria including elongation measures, with expert reports documenting LW values for challenged districts.

4.3 Threshold Interpretation

The LW > 3 threshold has appeared in litigation as follows:

- A district with LW $= 3$ is three times as long as it is wide—an aspect ratio that is difficult to justify on geographic grounds unless the district follows a natural geographic corridor (e.g., a river valley).
- LW > 4 suggests an extreme elongation that courts have found characteristic of "finger" or "tentacle" districts designed to capture distant population clusters.
- LW ≤ 2.5 is generally regarded as not elongated; values between 2.5 and 3.0 are contested.

4.4 Caveat: No Federal Legal Standard

No federal court has established LW > 3 as a binding legal threshold. The standard appears in expert reports and state court decisions as a heuristic, not a categorical rule. Practitioners should present LW as one component of a multi-metric compactness profile (K.7) rather than as a pass-fail criterion.

5 Empirical Comparison Across Structure Algorithms

5.1 Key Results

All bisect algorithms satisfy LW $\leq$ 2.8. The maximum LW across all 12 algorithm-state combinations is $2.79^{\dagger}$ (TX standard-bisect), comfortably below the LW > 3 court threshold. Moving-knife consistently produces the lowest maximum LW across all states.

Table 1: District-level LW (MBR) statistics by structure algorithm and state (2020 Census, seed = 42†). Note: lower LW is more compact; the court threshold is LW > 3.

State	Algorithm	Min	Median	Mean	Max
NC ($k = 14$)	standard-bisect	1.12	1.71	1.79†	2.68
	prime-factor	1.10	1.65	1.72†	2.54
	ratio-optimal	1.09	1.61	1.68†	2.43
	moving-knife	1.08	1.58	1.65†	2.38
WI ($k = 8$)	standard-bisect	1.10	1.56	1.61†	2.31
	prime-factor	1.09	1.51	1.55†	2.18
	ratio-optimal	1.08	1.47	1.51†	2.06
	moving-knife	1.07	1.44	1.48†	1.98
TX ($k = 38$)	standard-bisect	1.14	1.84	1.92†	2.79
	prime-factor	1.12	1.78	1.85†	2.67
	ratio-optimal	1.11	1.72	1.79†	2.58
	moving-knife	1.10	1.68	1.75†	2.51

†Single-run result, seed = 42. Maximum LW across all algorithm-state pairs is 2.79† (TX standard-bisect), below the LW > 3 court threshold.

NC under moving-knife. NC moving-knife achieves maximum LW = 2.38†, the best performance for NC. This reflects MKA's design: by maximising minimum Reock, it also controls elongation, since round (high Reock) districts tend to have low LW.

Zero districts exceed the threshold. Across all $4 \times 3 = 12$ algorithm-state combinations, zero districts exceed LW = 3.0. This is a strong result for litigation: a practitioner can certify that no bisect-generated district is "elongated" by the court-applied standard.

6 AABB vs. MBR Discrepancy Quantification

We quantify the AABB vs. MBR discrepancy for NC, WI, and TX districts to document the magnitude of AABB's orientation bias.

6.1 Discrepancy Measure

We define the AABB overestimate ratio:

$$\mathrm{Disc}(D) = \frac{\mathrm{LW}_{\mathrm{AABB}}(D)}{\mathrm{LW}_{\mathrm{MBR}}(D)} - 1,$$

so Disc = 0 means AABB and MBR agree; Disc = 0.41 means AABB overestimates LW by 41%.

6.2 Empirical Discrepancy

Table 2: AABB vs. MBR LW discrepancy statistics (ratio-optimal, seed = 42†).

State	Mean Disc	Max Disc	Median Disc
NC ($k = 14$)	0.12†	0.38†	0.09†
WI ($k = 8$)	0.09†	0.31†	0.07†
TX ($k = 38$)	0.14†	0.41†	0.11†

†Single-run result, seed = 42. Positive Disc means AABB overestimates LW (AABB LW > MBR LW).

The maximum discrepancy of 0.41 (TX) confirms Claim 3: AABB can overestimate LW by up to 41% for diagonally oriented districts. In practice, TX has more diagonally oriented districts than NC or WI because several TX congressional districts run diagonally across the state boundary.

6.3 When AABB Underestimates LW

In rare cases, AABB can underestimate LW (Disc < 0). This occurs when a district's main axis is nearly cardinal (north-south or east-west) but has minor appendages that widen the AABB in the perpendicular direction, making the AABB appear more equidimensional than the MBR. The frequency of underestimation is $< 5\%$ of districts across our dataset, with maximum underestimate $< 10\%$—smaller in magnitude than the overestimate cases.

6.4 Recommendation

AABB-based LW is *not legally defensible* because it is orientation-dependent. A court or opposing expert can challenge any AABB-based LW value by rotating the district geometry and obtaining a different result. Only MBR-based LW (rotating calipers) is rotation-invariant and thus legally credible.

7 Practitioner Usage Guide

LW provides a uniquely intuitive compactness metric for courtroom communication.

7.1 When to Use LW

- **Primary use**: When the legal challenge involves "elongation" claims—districts that run as corridors across the state.
- **Secondary use**: As a complement to Reock in the composite profile, because both measure elongation but from different geometric primitives (MBC vs. MBR).
- **Threshold certification**: When the jurisdiction applies a LW > 3 threshold (Illinois, New York expert practice), use LW as the primary pass-fail metric.

7.2 Courtroom Explanation

> *The Length-Width Ratio measures the aspect ratio of the smallest rectangle that can contain the district, oriented at any angle. A district with LW* $= 3$ *is three times as long as it is wide. All 14 NC congressional districts in the proposed plan have LW* ≤ 2.38*, well below the 3:1 threshold that courts have applied in redistricting challenges.*

7.3 Disclosure Statement

For court filings:

> *All Length-Width Ratio scores were computed using the minimum-area bounding rectangle (MBR) of district convex hulls, projected to Albers Equal Area Conic (EPSG:5070). The MBR was computed using the rotating calipers algorithm applied to the convex hull vertices. LW* $= \ell/w$ *where* $\ell \geq w$ *are the MBR dimensions. Axis-aligned bounding boxes (AABB) were not used because they are orientation-dependent.*

7.4 CLI Usage

```
bisect label-analyze <label> --year 2020 --types compactness
```

The resulting `compactness.json` includes the MBR-based `length_width_ratio` field for each district and `mean_length_width_ratio` in the summary. AABB values are diagnostic only: use `axis_aligned_length_width_ratio()` in `bisect-analysis` for orientation-bias checks, not as the production compactness score.

8 Conclusion

The Length-Width Ratio provides a visually intuitive elongation metric that complements the perimeter-based and area-based metrics in the K-track compactness framework. Our principal findings are:

1. The LW > 3 court threshold—applied in Illinois and New York redistricting litigation—represents a well-established elongation criterion that bisect plans satisfy by construction.

2. All four bisect structure algorithms produce LW $\leq 2.8^{\dagger}$ on NC 2020, with zero districts exceeding the LW > 3 threshold across all algorithm-state pairs studied.

3. AABB-based LW can overestimate true LW by up to 41% for diagonally oriented districts; only MBR-based LW (rotating calipers) is rotation-invariant and legally defensible.

Practitioners should use MBR-based LW computed via rotating calipers, explicitly exclude AABB from their methodology, and present LW alongside PP, Reock, and CH in the composite compactness profile (K.7). The ≤ 2.8 maximum LW for all bisect NC plans provides a strong certified result for litigation contexts where elongation is at issue. The implementation boundary is now explicit: BISECT's production compactness analysis reports only the MBR-based LW score, while AABB is retained as a tested diagnostic helper for explaining why orientation-dependent alternatives are not suitable for court-facing compactness reports.

References

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